

CSL351 : COMPUTER NETWORKS
Assignment - 11

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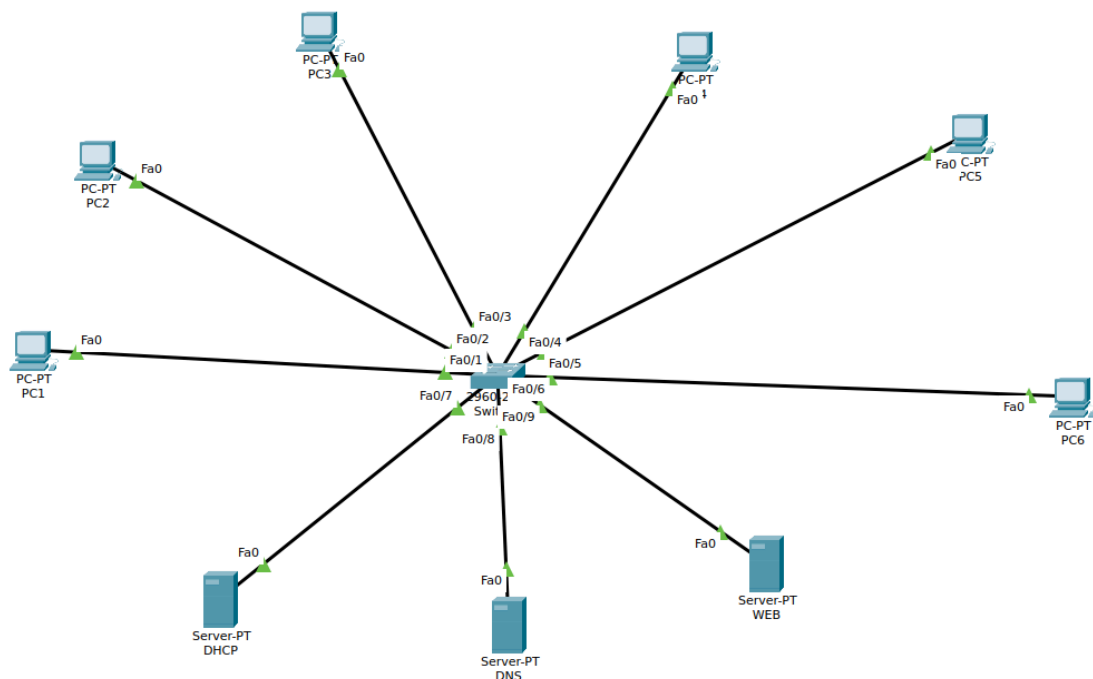
Google Drive Link :

<https://drive.google.com/drive/folders/1S49xoJARZZ1LACHAliyr1eMkgAMjm4if?usp=sharing>

Contains the video for the simulation and the assignment pkt file.

Report from Next Page.

Part A :

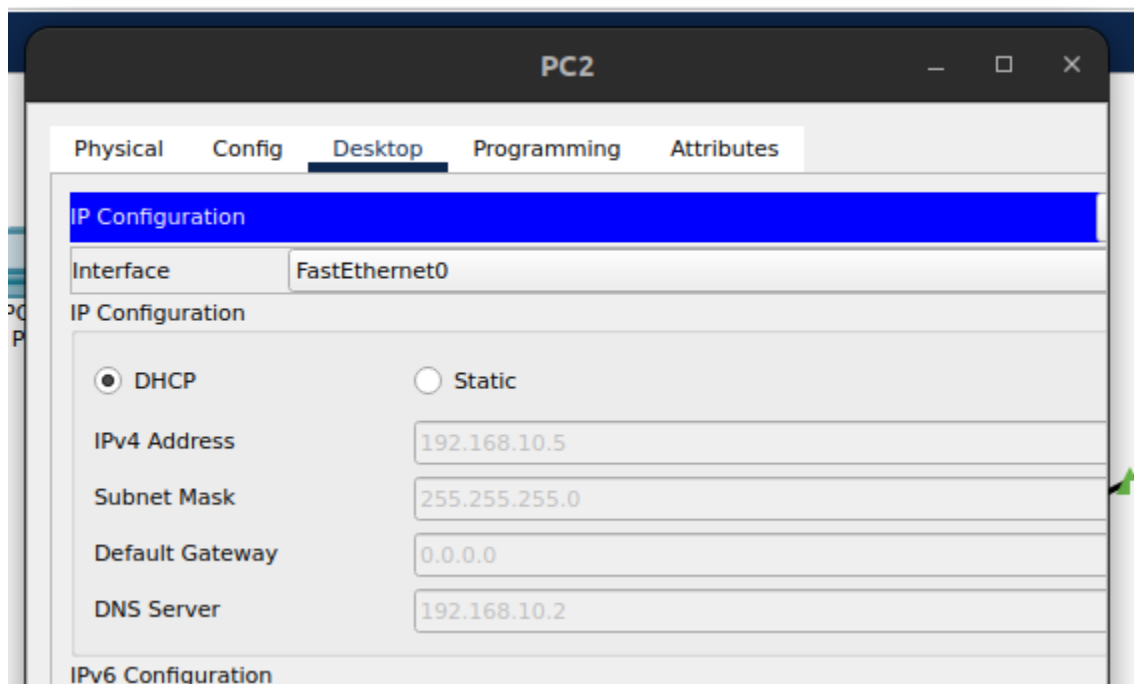
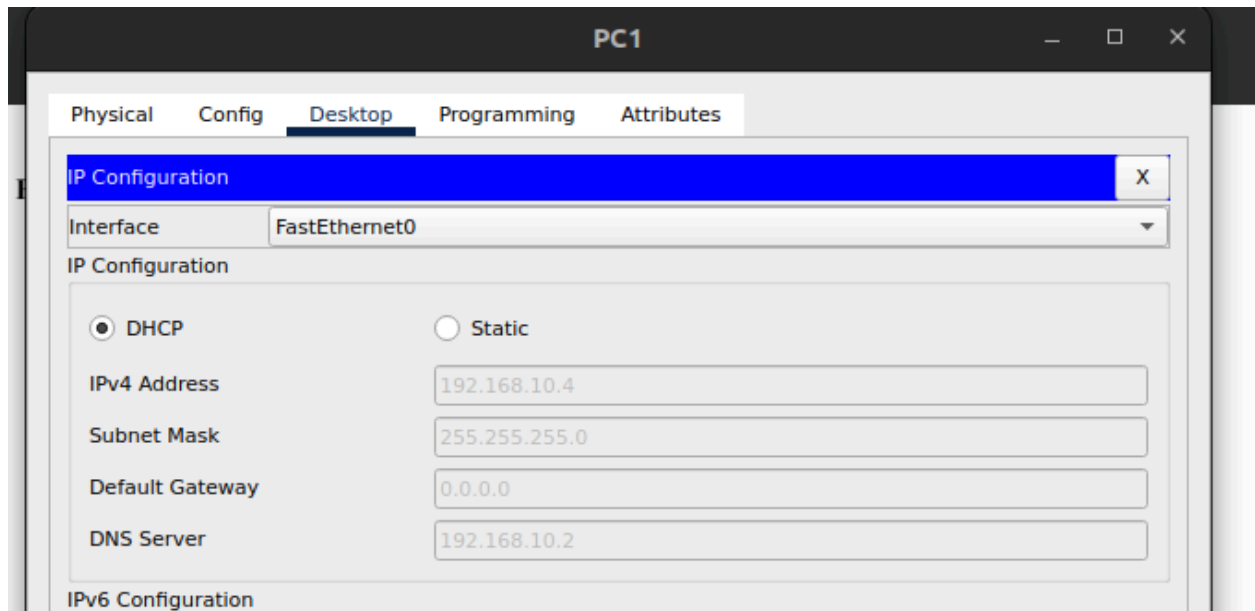


Explanation :

I have used the 2960 Switch which is connected to the 6 PCs via copper straight through cable and the 3 servers via the same copper straight through cable as specified by the question. For the servers i have used server-PT namely DHCP server, DNS server, WEB server. The Pcs i have used PC-PT named as (PC1,PC2 ,PC3 ... PC6). The interfaces used for connecting any two device is the fastethernet.

Part B :

1. PCs via DHCP



Explanation : Similarly i have configured all the 6PCs via enabling the DHCP and the DHCP server assigned them the IP address starting from 192.168.10.4 as fixed by me.

2. Server :

DHCP Server :

The screenshot shows a network configuration window titled "DHCP". The "Config" tab is selected, showing settings for the "FastEthernet0" interface. The left sidebar has a tree view with "GLOBAL" and "INTERFACE" sections. Under "INTERFACE", "FastEthernet0" is selected. The main area displays the following configuration:

- Port Status:** ☒ On
- Link Speed:** ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex:** ☐ Half Duplex ☒ Full Duplex ☒ Auto
- MAC Address:** 00D0.5830.D1DC
- IP Configuration:**
 - ☐ DHCP
 - ☒ Static
- IPv4 Address:** 192.168.10.1
- Subnet Mask:** 255.255.255.0
- IPv6 Configuration:**
 - ☐ Automatic
 - ☒ Static
- IPv6 Address:** [Empty field]
- Link Local Address:** FE80::2D0:58FF:FE30:D1DC

At the bottom left, there is a "Top" button.

Explanation : For the DHCP server i have used static ip address which is 192.168.10.1

DNS Server :

The screenshot shows a network configuration window titled "DNS" with a dark header bar. Below the header, there are tabs: "Physical", "Config" (which is highlighted with a blue underline), "Services", "Desktop", "Programming", and "Attributes". On the left side, there is a sidebar with a tree view containing "GLOBAL" (with sub-items "Settings" and "Algorithm Settings") and "INTERFACE" (with sub-item "FastEthernet0"). The main area of the window is titled "FastEthernet0" and contains the following configuration options:

- Port Status:** ☒ On
- Link Speed:** ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex:** ☐ Half Duplex ☒ Full Duplex ☒ Auto
- MAC Address:** 00D0.977B.D2A6
- IP Configuration:** ☐ DHCP ☒ Static
 - IPv4 Address:** 192.168.10.2
 - Subnet Mask:** 255.255.255.0
- IPv6 Configuration:** ☐ Automatic ☒ Static
 - IPv6 Address:** [Empty field] / [Empty field]
 - Link Local Address:** FE80::2D0:97FF:FE7B:D2A6

At the bottom left of the window, there is a "Top" button with a small square icon next to it.

Explanation : For the DNS server i have used static ip address which is 192.168.10.2

WEB Server :

The screenshot shows a web browser window titled "WEB" with a navigation bar containing "Physical", "Config", "Services", "Desktop", "Programming", and "Attributes". The "Config" tab is active. On the left, a sidebar menu has "GLOBAL" (with sub-items "Settings" and "Algorithm Settings") and "INTERFACE" (with "FastEthernet0" selected). The main content area is titled "FastEthernet0" and contains the following settings:

- Port Status: ☒ On
- Link Speed: ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto
- MAC Address: 0001.42AC.63D5
- IP Configuration:
 - ☐ DHCP
 - ☒ Static
 - IPv4 Address: 192.168.10.3
 - Subnet Mask: 255.255.255.0
- IPv6 Configuration:
 - ☐ Automatic
 - ☒ Static
 - IPv6 Address: [empty field] / [empty field]
 - Link Local Address: FE80::201:42FF:FEAC:63D5

At the bottom left, there is a "Top" link with a small square icon next to it.

Explanation : For the WEB server i have used static ip address which is 192.168.10.3

3. Name and IP mapping

The screenshot shows a web-based configuration interface for DNS. The top navigation bar includes tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'Services' tab is selected. On the left, a 'SERVICES' list contains various services, with 'DNS' highlighted. The main area is titled 'DNS' and contains the following elements:

- DNS Service:** A toggle switch set to 'On'.
- Resource Records:** A section for adding and managing DNS records.
- Name:** A text input field containing 'www.ashish.com'.
- Type:** A dropdown menu set to 'A Record'.
- Address:** A text input field containing '192.168.10.3'.
- Buttons:** 'Add', 'Save', and 'Remove' buttons.
- Table:** A table listing the configured resource records.
- DNS Cache:** A button at the bottom of the resource records section.

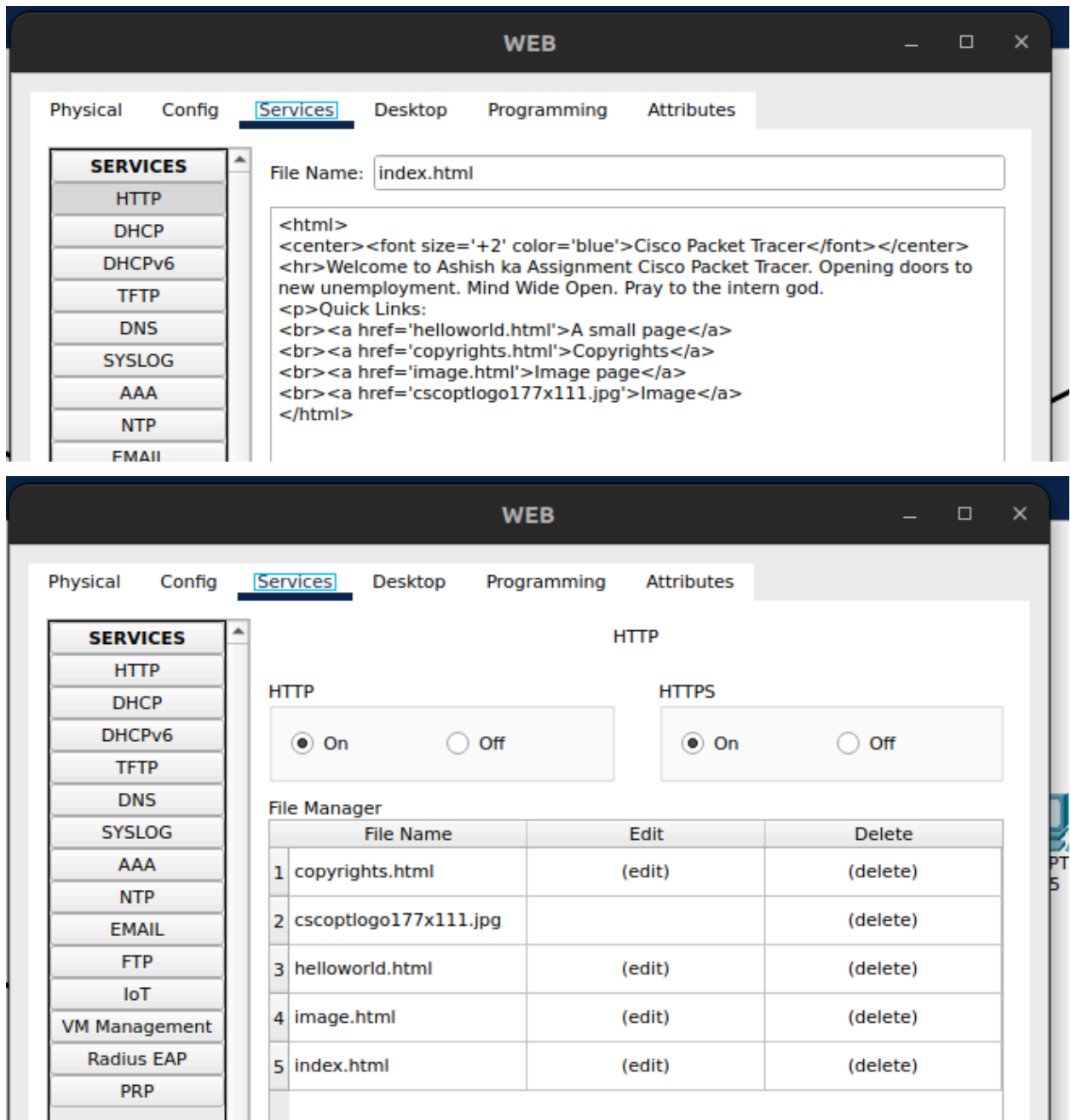
No.	Name	Type	Detail
0	www.ashish.com	A Record	192.168.10.3

At the bottom left, there is a 'Top' link.

Explanation : For the name and ip mapping i have called my website “www.ashish.com” and the ip address which is written is the IP of web server as specified above which is inserted via the type A record

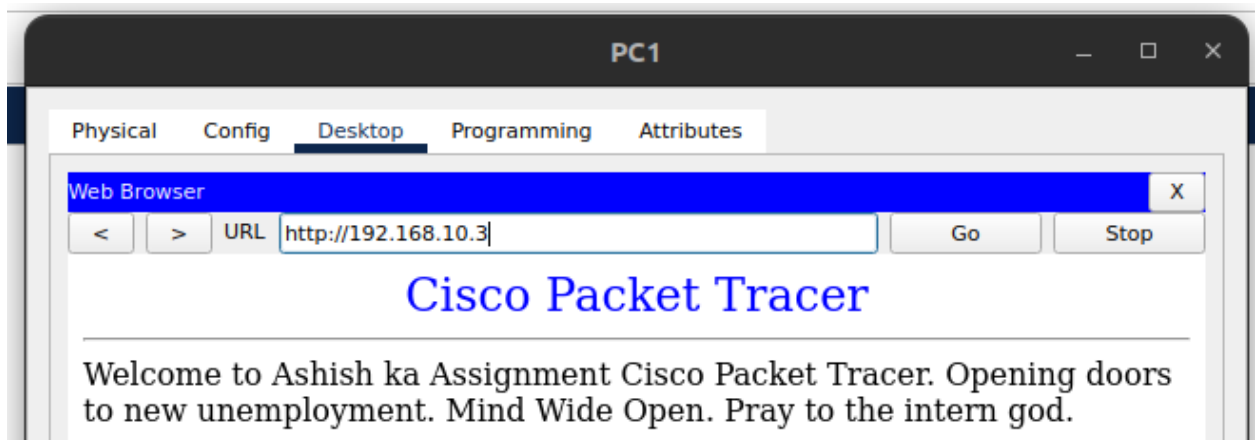
4. Web Server Demo

Image showing both the edited content of index.html and the HTTP and HTTPS service being on :



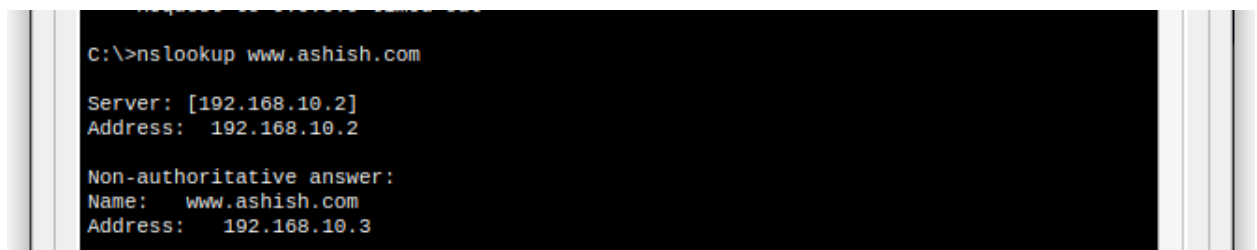
Explanation : I have modified the [index.html](#) given by the cisco packet tracer with some minor changes in it, and the the HTTP and HTTPS are enabled for the web server.

Image shows both the IP and the name of the server works correctly :



COMMAND USED :

nslookup www.ashish.com

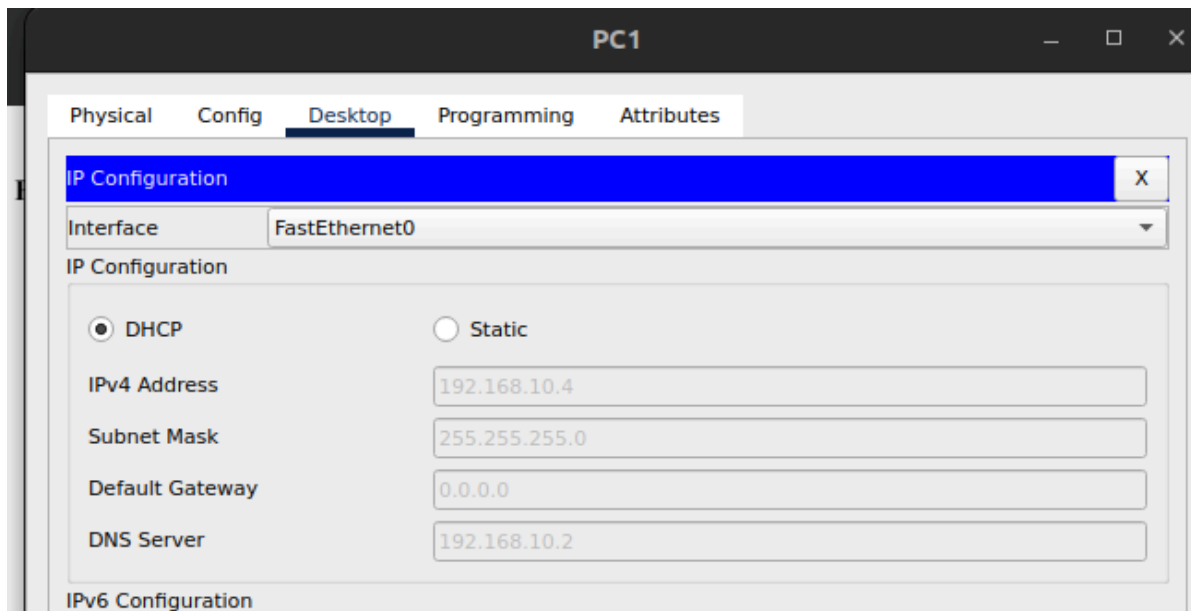


Explanation : I have tried to access the "www.ashish.com" via PC1 which successfully shows the content modified by me, which can be accessed via the IP addr of the web server or the URL of the web server. I have also used nslookup on that website to show that DNS correctly resolves the name to the corresponding web server properly.

Part C :

1. PCs IP config via DHCP :

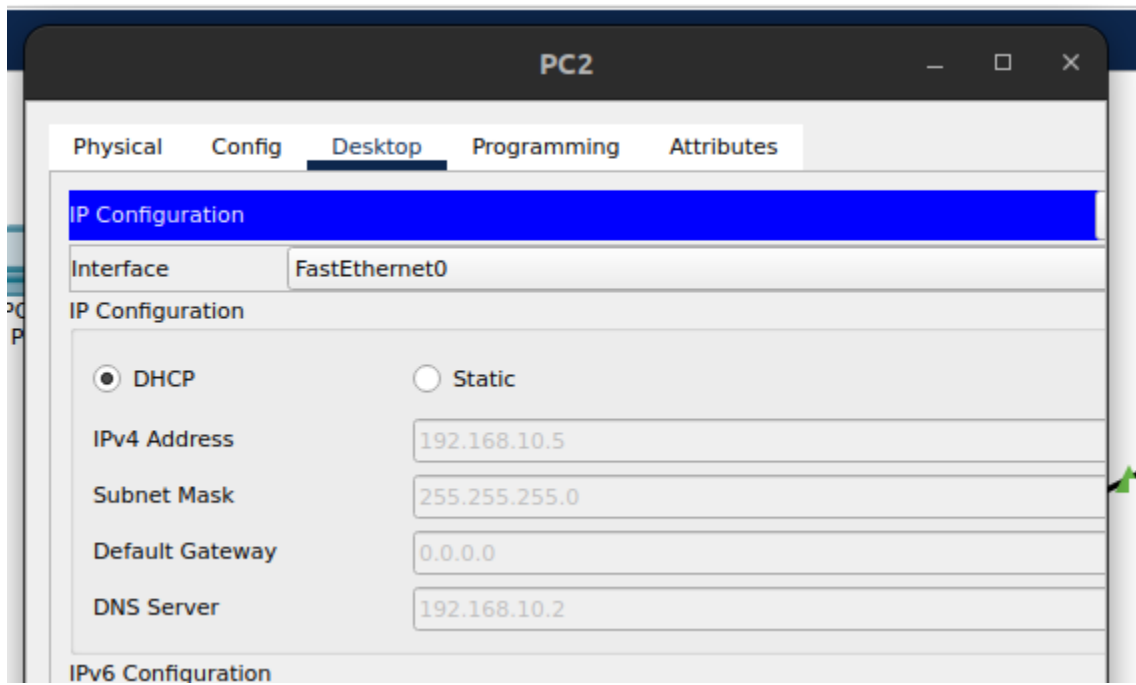
a. PC1



The screenshot shows the configuration window for PC1. The 'Desktop' tab is selected, and the 'IP Configuration' section is highlighted. The interface is set to 'FastEthernet0'. The 'DHCP' radio button is selected, and the 'Static' radio button is unselected. The IPv4 Address is set to 192.168.10.4, the Subnet Mask is 255.255.255.0, the Default Gateway is 0.0.0.0, and the DNS Server is 192.168.10.2. The IPv6 Configuration section is visible below the IPv4 Configuration section.

Interface	FastEthernet0
IPv4 Address	192.168.10.4
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	192.168.10.2

b. PC2



The screenshot shows the configuration window for PC2. The 'Desktop' tab is selected, and the 'IP Configuration' section is highlighted. The interface is set to 'FastEthernet0'. The 'DHCP' radio button is selected, and the 'Static' radio button is unselected. The IPv4 Address is set to 192.168.10.5, the Subnet Mask is 255.255.255.0, the Default Gateway is 0.0.0.0, and the DNS Server is 192.168.10.2. The IPv6 Configuration section is visible below the IPv4 Configuration section.

Interface	FastEthernet0
IPv4 Address	192.168.10.5
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	192.168.10.2

c. PC3

The screenshot shows the configuration window for PC3. The 'Desktop' tab is selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. The 'IP Configuration' section has two radio buttons: 'DHCP' (selected) and 'Static'. Below these are five text input fields: 'IPv4 Address' (192.168.10.6), 'Subnet Mask' (255.255.255.0), 'Default Gateway' (0.0.0.0), and 'DNS Server' (192.168.10.2). The 'IPv6 Configuration' section is partially visible at the bottom.

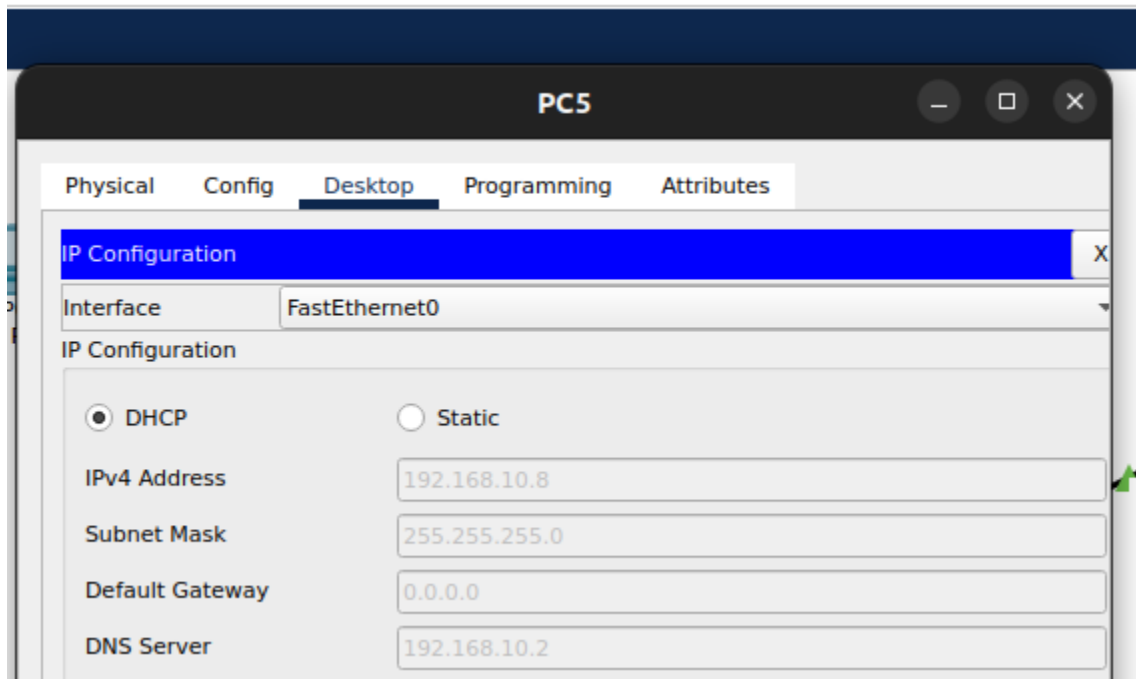
Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.10.6
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	192.168.10.2
IPv6 Configuration	

d. PC4

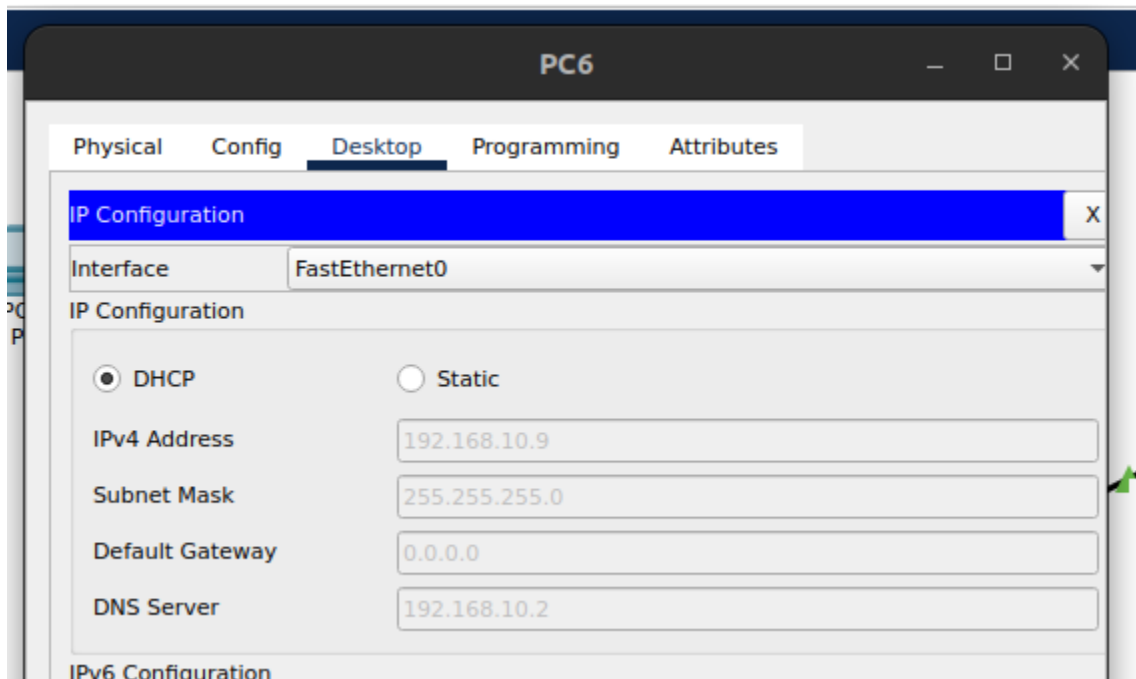
The screenshot shows the configuration window for PC4. The 'Desktop' tab is selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. The 'IP Configuration' section has two radio buttons: 'DHCP' (selected) and 'Static'. Below these are five text input fields: 'IPv4 Address' (192.168.10.7), 'Subnet Mask' (255.255.255.0), 'Default Gateway' (0.0.0.0), and 'DNS Server' (192.168.10.2). The 'IPv6 Configuration' section is partially visible at the bottom.

Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.10.7
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	192.168.10.2
IPv6 Configuration	

e. PC5



f. PC6

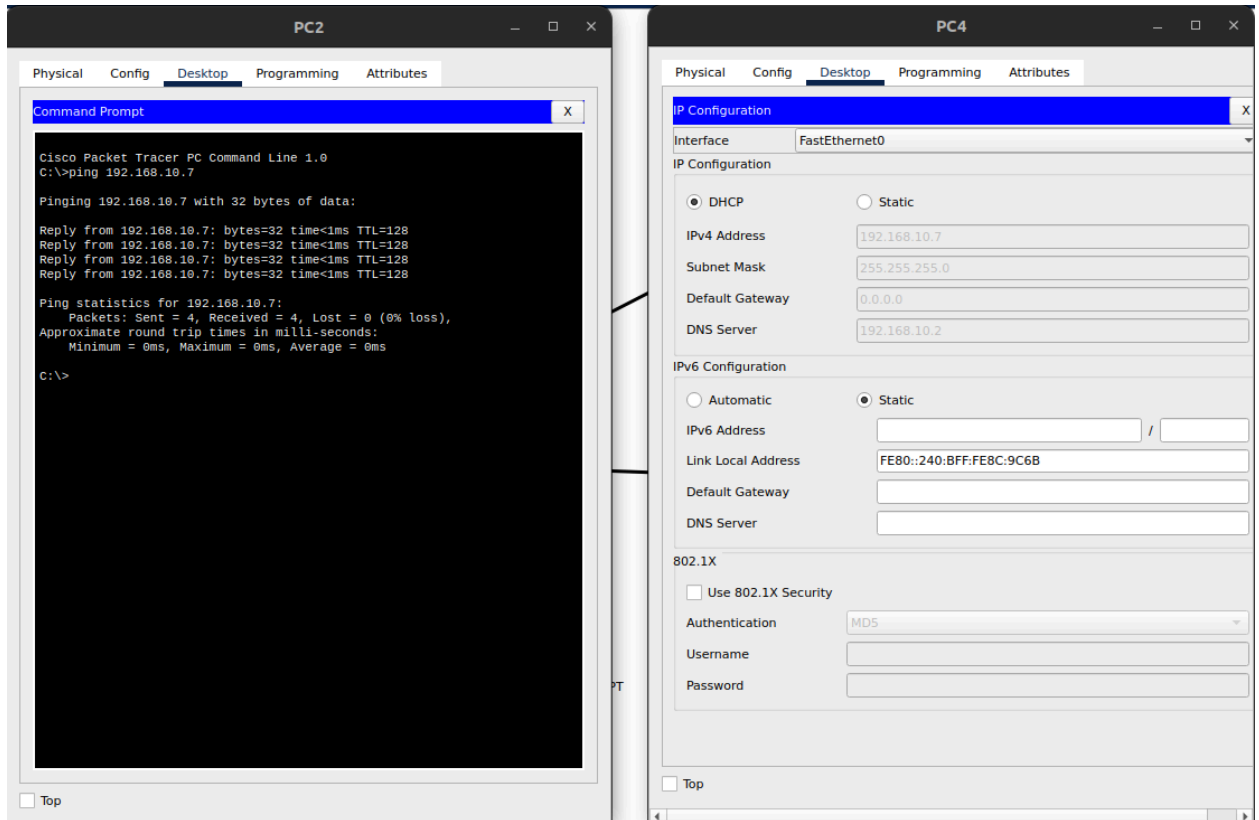


Explanation : I have configured all the 6PCs via enabling the DHCP and the DHCP server assigned them the IP address starting from 192.168.10.4 as fixed by me.

2. Ping between two PCs

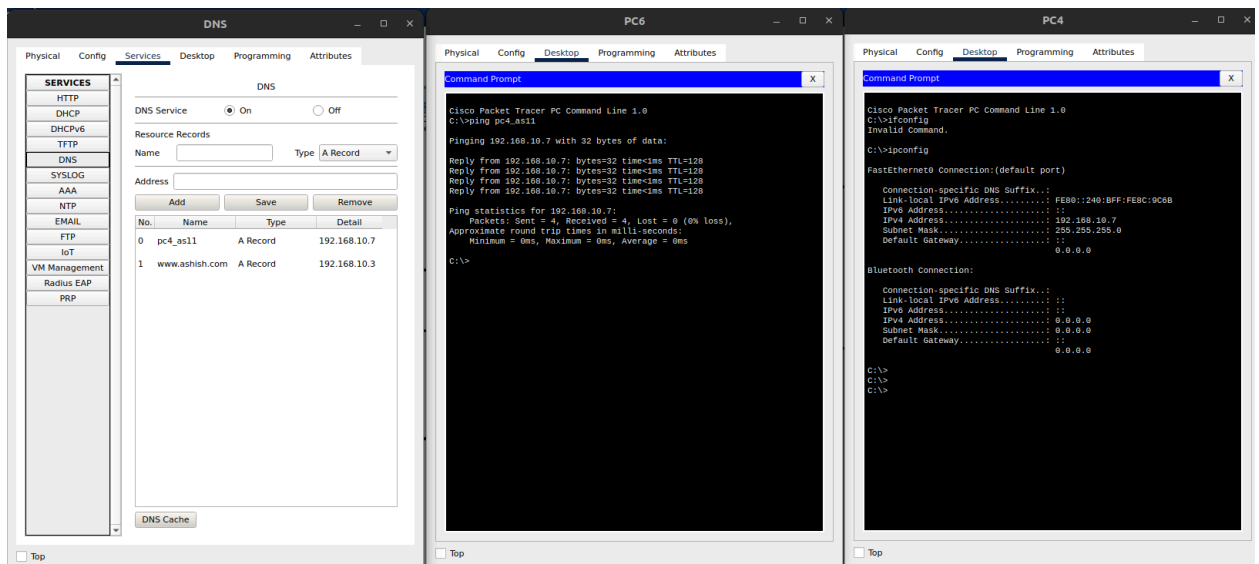
a. IP Address

pinging pc 4 from pc 2 :



b. Domain Name

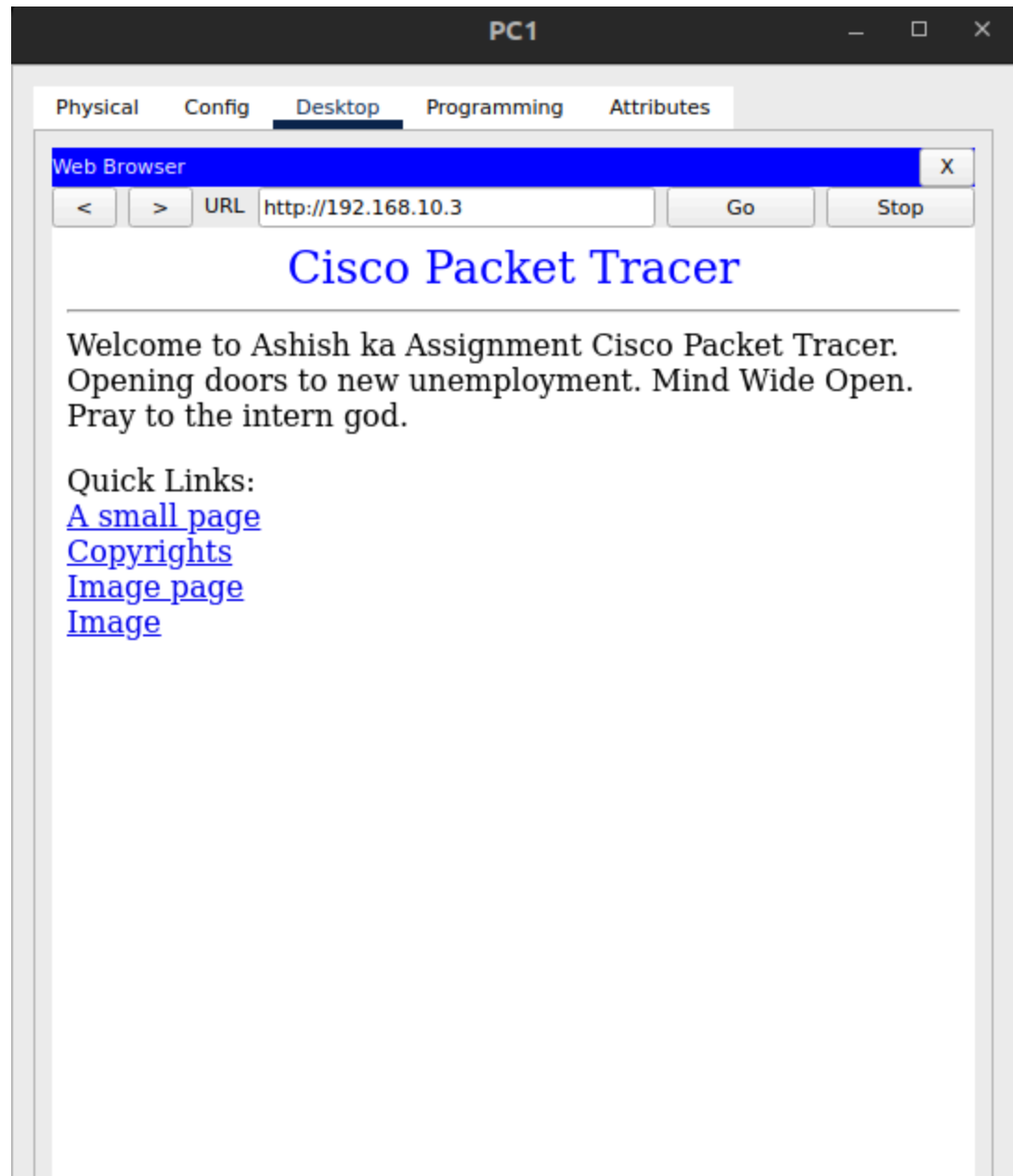
pinging pc4 from pc6



Explanation : For pingging via IP address i did ipconfig on the other PC noted its IP and then did ping [that_ip] to show it correctly pings that website. For ping via domain name i have added the record for the corresponding ip in the DNS server and then ping using that name.

3. Access the web server :

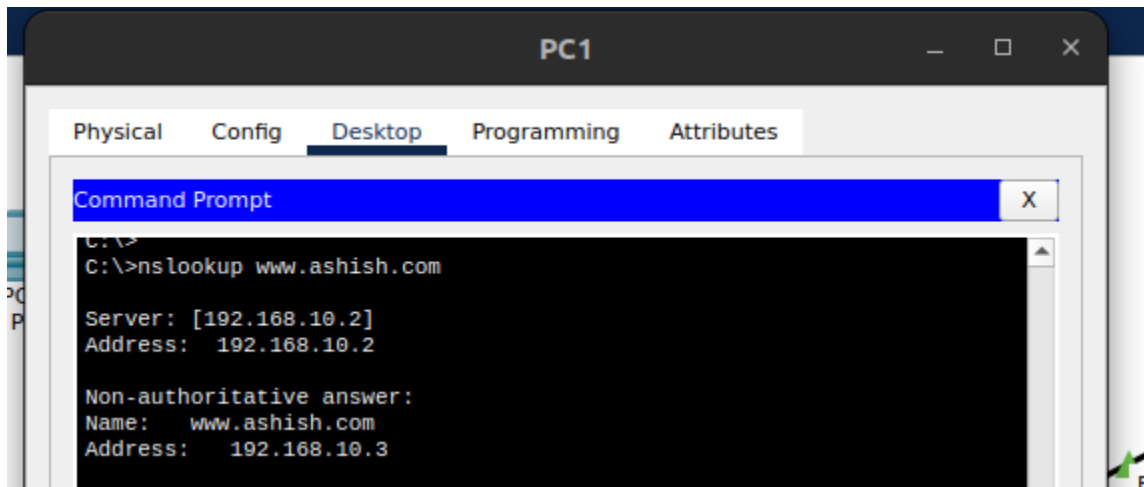
a. IP Address



b. Domain Name



4. DNS resolution working :



The screenshot shows a virtual machine window titled 'PC1'. Inside, there are tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the command 'C:\>nslookup www.ashish.com' and its output: 'Server: [192.168.10.2]', 'Address: 192.168.10.2', 'Non-authoritative answer:', 'Name: www.ashish.com', and 'Address: 192.168.10.3'.

```
C:\>nslookup www.ashish.com

Server: [192.168.10.2]
Address: 192.168.10.2

Non-authoritative answer:
Name: www.ashish.com
Address: 192.168.10.3
```

Explanation : This demonstrates the DNS resolution is working as it perfectly shows the IP address resolving for that url. This can be seen in the video provided more clearly in the Google Drive Link.